

Lab 4, Act 3

The model is wrong. Does it matter? (optional)

NRES 746 – name: _____ group: _____

Picking up where we left off

This Act is optional. It's extra practice connecting Act 1's method to a question every real analyst eventually faces: what happens if you get the model wrong? It isn't required for Lab 4, but Act 2's whole power analysis leaned on the same log-scale method you'll interrogate here, so it's worth doing if you have time.

In Act 1 you fit a straight line to the *log* counts, which matches how the data were actually generated, and your simulations confirmed the slope estimate was unbiased with a spread that matched a formula you checked it against. Real analysts do not get to pick the analysis that matches the truth. Suppose a second analyst, following the agency's old spreadsheet protocol, fits a straight line to the *raw* counts instead. That model says the index falls by the same *number* of animals every year. The truth says it falls by the same *proportion*.

We will compare the two analysts under two scenarios that differ only in how fast the population is declining. Both scenarios follow the *same* data-generating model as Act 1: lognormal counts, a log-linear decline, and constant scatter σ on the log scale, with $I_0 = 60$ and $\sigma = 0.2$ throughout, and annual surveys ($t = 1, \dots, 25$). Only the rate of decline, b , differs:

Scenario	Total decline, year 0 to year 25
Mild	25% over 25 years
Severe	90% over 25 years

Both analysts report the same management quantity: the estimated **percent decline over the 25 years**. The log-scale analyst computes it as $1 - e^{25b}$. The raw-scale analyst computes it as $1 - \hat{I}_{25}/\hat{I}_0$, using the fitted values of the straight line at $t = 25$ and $t = 0$.

Does the wrong model give the wrong answer? And can the data alone tell us the model is wrong?

Try Part 1 on paper first, before you open a laptop or start typing code. Resist the urge to jump straight to R or an AI assistant.

Part 1: What does each model assume about scatter? (individual, ~10 min)

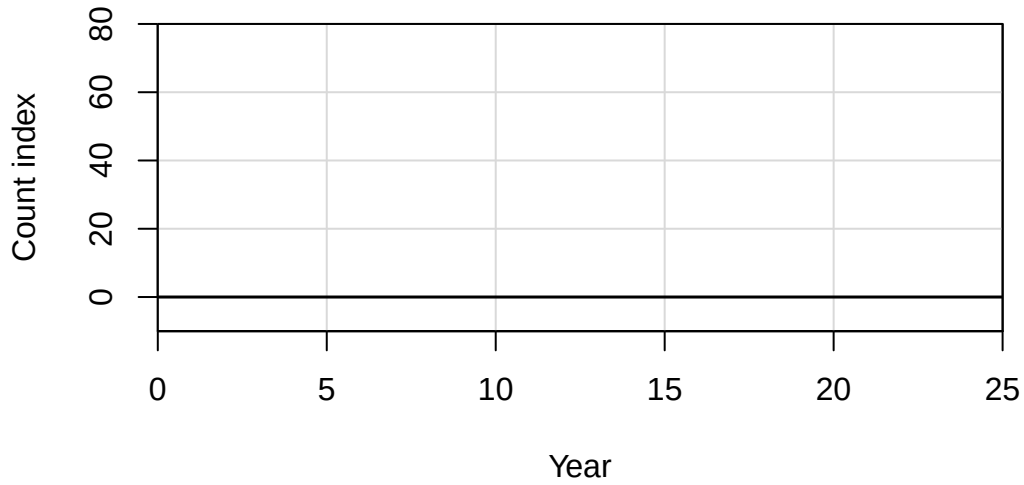
1a. The *variance function* of a distribution says how the variance changes with the mean μ . Fill in the blanks. A gamma distribution with shape α and scale s has mean αs and variance αs^2 . Bolker Ch. 4 is fair game.

Distribution	Var(Y) in terms of the mean	Constant CV?
Normal (SD sigma)		
Poisson		
Negative binomial (mean, size k)		
Gamma (mean, shape alpha)		
Lognormal (log-SD s)	$(\exp(s^2) - 1) \times \text{mean}^2$	

1b. For a lognormal variable whose log has mean m and SD s , the mean of the variable is $e^{m+s^2/2}$ and its SD is $\sqrt{e^{s^2} - 1}$ times that mean. In our model $s = \sigma = 0.2$ and the median in year 0 is $e^m = 60$. Fill in the table (using a calculator or your phone as a calculator is fine).

Year and scenario	Median	Mean	SD
Year 0	60		
Year 25, mild	45		
Year 25, severe	6		

1c. For the severe scenario, use the axes below to sketch (i) the median trajectory, (ii) a band showing roughly ± 2 SD around it, and (iii) the straight line an analyst would fit through the counts. Does this visualization point to any logical flaws with the straight line model?



Part 2: Commit to predictions (individual, ~4 min)

2a. For each analyst and scenario, will the estimated percent decline be unbiased, too high, or too low? Write a word or two in each cell. Then say briefly why.

	Mild (25%)	Severe (90%)
Log-scale analyst		
Raw-scale analyst		

Part 3: A goodness of fit check (individual, ~5 min)

In real life we never know the true model. What we can do is ask whether the data look like they could have come from the *fitted* model: fit the model, simulate many new datasets from the fitted model, and compare a summary statistic of the real data with the same statistic across the simulated datasets.

Our statistic is the number of **sign changes** in the residuals over the 25 years (the number of times neighboring residuals switch between positive and negative). If the fitted line captures the pattern and the residuals are independent, about half of the 24 neighboring pairs should switch. If the true pattern is curved and the line is straight, residuals come in long runs of one sign, and very few switch.

3a. Write, in plain English, the procedure for a check that returns a p -value: the fraction of simulated datasets with as few sign changes as the real data. Do not write R code. Be specific about what you fit to each simulated dataset and why.

3b. How often do you expect this check to flag each model in each scenario (rarely, sometimes, or usually)?

	Mild (25%)	Severe (90%)
Log-scale model		
Raw-scale model		

Checkpoint: finalize your plan (~5 min)

Reread your procedure from 3a and your predictions from 2a and 3b. Would your plan actually work if you started typing it up right now? Revise anything incomplete or unclear before you move on – this is your last checkpoint before writing any code.

Laptops open now. Everything from here on is coding. You're welcome to use an AI assistant throughout – record what you asked for and what you had to fix, the same way Act 2 asked you to.

Part 4: Run the experiments (~40 min)

Use these settings: $I_0 = 60$, $\sigma = 0.2$, $t = 1, \dots, 25$, $b_{\text{mild}} = \ln(0.75)/25$ and $b_{\text{severe}} = \ln(0.10)/25$. The two helper functions below are ready-made. Type them into your R script.

```
# number of times neighboring residuals switch sign
sign_changes <- function(r) {
  s <- sign(r)
  sum(s[-1] != s[-length(s)])
}

# t: years, y: observed counts, sims: matrix of simulated counts
# (one row per year, one column per simulated data set)
plot_envelope <- function(t, y, sims) {
  lo <- apply(sims, 1, quantile, 0.05)
  hi <- apply(sims, 1, quantile, 0.95)
  plot(t, y, type = "n", ylim = range(y, lo, hi), xlab = "Year", ylab = "Count index")
  polygon(c(t, rev(t)), c(lo, rev(hi)), col = "grey85", border = NA)
  points(t, y, pch = 16)
  abline(h = 0, lty = 3)
}
```

The three experiments:

1. **Two analysts.** For each scenario, simulate 2,000 datasets from the model. Fit both analysts to every dataset and compute each one's estimated percent decline. Record the mean and SD of the estimates and the share of estimates above 100%.
2. **An envelope.** Simulate one dataset from the severe scenario and fit the raw-scale model. Simulate 500 new datasets from the fitted raw-scale model (Normal with mean equal to the fitted values and SD equal to the residual SD). Draw the envelope with `plot_envelope()`.
3. **The check.** Implement your procedure from 3a as a function. For the log-scale model, run the same check on $\log(y)$, with the simulated datasets also generated on the log scale. Use 200 simulated datasets inside the check. Apply it to 100 fresh datasets for each scenario and each model, and record how often it returns $p < 0.05$.

Write the code for your three experiments (R's built-in help, documentation, human-written websites, an AI assistant, and your own notes from Act 2 are all fair game).

Notes on how you approached this (if you used an AI assistant, record your prompt and what you had to change to get it to do what you intended; if not, note what gave you the most trouble):

Results

Experiment 1. Estimated percent decline over 25 years.

Scenario	Analyst	Mean	SD	Share > 100%
Mild (25%)	Log-scale			
Mild (25%)	Raw-scale			
Severe (90%)	Log-scale			
Severe (90%)	Raw-scale			

Experiment 3. Share of datasets for which the check returns $p < 0.05$.

	Mild (25%)	Severe (90%)
Log-scale model		
Raw-scale model		

In one or two sentences, describe what the envelope in Experiment 2 shows that a single glance at the raw counts might not.

If you want to go even further: simulate 100,000 values of I_0 from the model. Compare `mean(log(i0))` with `log(mean(i0))`. The gap should match $\sigma^2/2$ from 1b. Why does the average of the logs differ from the log of the average?

Reflection (~5 min)

How did your predictions in 2a and 3b compare with your results? In which scenario did the wrong model matter, and could the check have warned you?

Please take a minute to reflect on any areas of confusion (“muddiest points”) or topics you want to review or dust off.
